GLUCOSAMINE + CHONDROITIN

Glucosamine Sulfate 500mg • Chondroitin Sulfate 400mg

COMPOSITION:
Nuvanta Glucosamine+Chondroitin Capsule is a size 00EL colourless hard gelatin capsule filled with off-white coloured free flowing powder, containing Glucosamine sulfate potassium chloride complex 663.31mg and shark cartilage derived Chondroitin sulfate 425.54mg, equivalent to Glucosamine sulfate 500mg and Chondroitin sulfate 400mg.

EACH TABLET CONTAINS:

Active Ingredients	Label Claim
Glucosamine Sulfate	500mg
Chondroitin Sulfate	400mg

INDICATIONS:
Nuvanta Glucosamine+Chondroitin Capsule is indicated as adjunctive therapy in osteoarthritis.

DOSAGE & ADMINISTRATION:
Tablets should be taken 15-30 minutes before meal.

MILD OR MODERATE OSTEOARTHRITIC SYMPTOMS:
1 capsule taken twice daily for at least 6 weeks or according to medical prescription.

SEVERE OSTEOARTHRITIC SYMPTOMS:

Initial Therapy: 1 capsule 3 times daily for at least 8-12 weeks.

Follow-Up Therapy: 1-2 capsules daily for additional 3-4 months.

The treatment of osteoarthritis should be repeated every 6 months or less as directed by medical professional.

Nuvanta Glucosamine+Chondroitin Capsule is an adjuvant therapy; the therapeutic effect may be seen after 1-2 weeks from initial therapy. If pain is severe, it's advisable to add on an anti-inflammatory drug for pain relieve at the beginning of treatment.

ROUTE OF ADMINISTRATION: Oral

PHARMACOLOGY & PHARMACODYNAMIC:
Glucosamine and Chondroitin play an important role in the promotion and maintenance of the structure and function of cartilage in the joints of the body. Glucosamine may also have anti-inflammatory properties.

Biochemically, glucosamine is involved in glycoprotein metabolism. Glycoproteins, known as proteoglycans, form the ground substance in the extra-cellular matrix of connective tissue. Proteoglycans are poly-anionic substances of high-molecular weight and contain many different types of hetero-polysaccharide side-chains covalently linked to a polypeptide-chain backbone. These polysaccharides make up to 95% of the proteoglycan structure. In fact, chemically, proteoglycans resemble polysaccharides more than they do proteins.

The polysaccharide groups in proteoglycans are called glycosaminoglycans or GAGs. GAGs include hyaluronic acid, chondroitin sulfate, dermatan sulfate, keratan sulfate, heparin and heparan sulfate. All of the GAGs contain derivatives of glucosamine or galactosamine. Glucosamine derivatives are found in hyaluronic acid, keratan sulfate and heparan sulfate. Chondroitin sulfate contains derivatives of galactosamine.

The glucosamine-containing glycosaminoglycan hyaluronic acid is vital for the function of articular cartilage. GAG chains are fundamental components of aggrecan found in articular cartilage. Aggrecan confers upon articular cartilage shock-absorbing properties. It does this by providing cartilage with a swelling pressure that is restrained by the tensile forces of collagen fibers. This balance confers upon articular cartilage the deformable resilience vital to its function. Chondroitin works in conjunction with Glucosamine.

It increases the production of synovial fluid (the fluid around the joint) attracting healthy nutrients, while adding increased elasticity and fluidity to the joints. It also reduces calcium salt deposits that can increase pain and inflammation, and blocks enzymes that can destroy cartilage.

In summary, Glucosamine and chondroitin stimulate the biosynthesis of mucopolysaccharides, which are the essential components of cartilage fundamental components. It also improves the synovial fluid viscosity and increases synovial fluid production.

These properties and actions contributed to the therapeutic effects of Glucosamine and Chondroitin in relieving osteoarthritic related symptoms such as:

- Relieves of articular aches and joint pains.
- Improves articular functions and increases joint flexibility and mobility.
- Prevent or delay the articular degenerative process.
- Nourishes connective tissue & aids in rebuilding of cartilages.

PHARMACOKINETIC:

Glucosamine

- **Absorption** - After oral administration, bioavailability is low due to first-pass hepatic metabolism ~ 26%. The gastrointestinal absorption is close to 90%.
- **Distribution** - Glucosamine is not protein-bound, but rather incorporates into plasma proteins (primarily globulins).
  Volume of Distribution: 2.5 Liters
- **Metabolism** - *Liver, Extensive*
  The first-pass effect in the liver in which more than 70% of glucosamine is metabolized.
- **Excretion** - *Renal Excretion - 10%, Feces - 11%*
  Part of a dose of glucosamine sulfate is eliminated as carbon dioxide via exhaled air.

Chondroitin
As for Chondroitin, its pharmacokinetic studies performed on humans and experimental animals after oral administration of chondroitin sulfate revealed that it can be absorbed orally. Chondroitin sulfate shows first-order kinetics up to single doses of 3,000 mg. Multiple doses of 800 mg in patients with osteoarthritis do not alter the kinetics of chondroitin sulfate. The bioavailability of chondroitin sulfate ranges from 15% to 24% of the orally administered dose. More particularly, on the articular tissue, Ronca et al reported that chondroitin sulfate is rapidly absorbed in the gastro-intestinal tract and a high content of labeled chondroitin sulfate is found in the synovial fluid and cartilage.

CONTRAINDICATION:

As with other glucosamine/chondroitin preparations, it's contraindicated in individuals who are hypersensitive to shellfish or shark cartilage. Patients with known hypersensitivity to any of its active component should avoid taking this preparation.

WARNING & PRECAUTION:

Glucosamine may increase insulin resistance. Glucosamine increases insulin resistance in normal and experimentally diabetic animals. In these animals, intravenous glucosamine significantly decreases the rate of glucose uptake in skeletal muscle. In animals given oral glucosamine, this is not observed. Those with type 2 diabetes and those who are overweight and have problems with glucose tolerance should have their blood sugars carefully monitored if they use glucosamine supplements.

Glucosamine/Chondroitin is not meant to replace NSAIDs, in cases of intense pain, as this is a causal therapy. It is advisable to take an anti-inflammatory drug in the initial treatment of osteoarthritis together with Glucosamine/Chondroitin.

Administration during the first three months of pregnancy must be avoided. Safety and effectiveness have not been established in children therefore, children should avoid using glucosamine.

The administration in patients with severe hepatic or renal insufficiency should be made under medical supervision. Derived from Seafood, therefore patients who are hypersensitive to shellfish should avoid taking this preparation.

A doctor should be consulted in order to exclude the presence of other joint conditions/diseases for which an alternative treatment should be considered.

Effects on Ability to Drive and Use Machines: No effects on the ability to drive or to operate machines are expected.

PREGNANCY & LACTATION:

As with other drugs, because of insufficient safety data, the use of glucosamine and chondroitin in children, pregnant women and nursing mothers should be given under medical supervision. Administration during the first 3 months of pregnancy must be avoided.

SIDE EFFECTS:

- **Cardiovascular:** Peripheral oedema, tachycardia were reported in a few patients following larger clinical trials investigating oral administration in osteoarthristis. Causal relationship has not been established.

- **Central Nervous System:** Drowsiness, headache, insomnia have been observed rarely during therapy (<1%).

- **Gastrointestinal:** Nausea, vomiting, diarrhea, dyspepsia or epigastric pain, constipation, heartburn and anorexia have been described rarely during oral therapy with glucosamine.

- **Skin:** Skin reactions such as erythema and pruritis have been reported with therapeutic administration of glucosamine.

INTERACTIONS WITH OTHER MEDICAMENTS

Effects on glucose metabolism & antidiabetic agents:

It has been hypothesized that glucosamine may impair insulin secretion through competitive inhibition of glucokinase in pancreatic beta cells and/or alteration of peripheral glucose uptake.

Glucosamine may increase insulin resistance and consequently affect glucose tolerance.

It may reduce antidiabetic agent effectiveness eg when used with these antidiabetic agent :

Acarbose, Acetohexamide, Chlorpropamide, Glipizidede, Glyburide, Metformin, Miglitol, Pioglitazone, Repaglinide, Rosiglitazone, Glimepiride, Tolbutamide, Troglitazone.

Glucosamine is likely safe in patients with well-controlled diabetes (HbA1c less than 6.5%) taking one or two oral antidiabetic medications or controlled by diet only. In patients with higher HbA1c levels or those taking insulin, monitor blood glucose levels closely.

Reduced effectiveness when used with glucosamine:

Doxorubicin, Etoposide, Teniposide

Warfarin

- Elevations of International Normalized Ratio serum values and potentiation of anticoagulant effects.

- If concomitant therapy is necessary, the patient's INR should be more closely monitored.

SYMPTOMS AND TREATMENT OF OVERDOSAGE:

No cases of accidental or intentional overdose are known. The animal acute and chronic toxicological studies indicate that toxic effects and symptoms of toxicity are not likely to occur, even after high overdoses.

Store below 30°C, away from direct sunlight.

Shelf life: 3 Years.

Pack Size: 10's (Sample), 20's (Sample), 30's, 50's, 60's, 90's
Not all pack sizes are marketed.

Reg. No.: MAL13085054XC

Keep out of reach of children.
Jauhkan daripada capaian kanak-kanak.

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